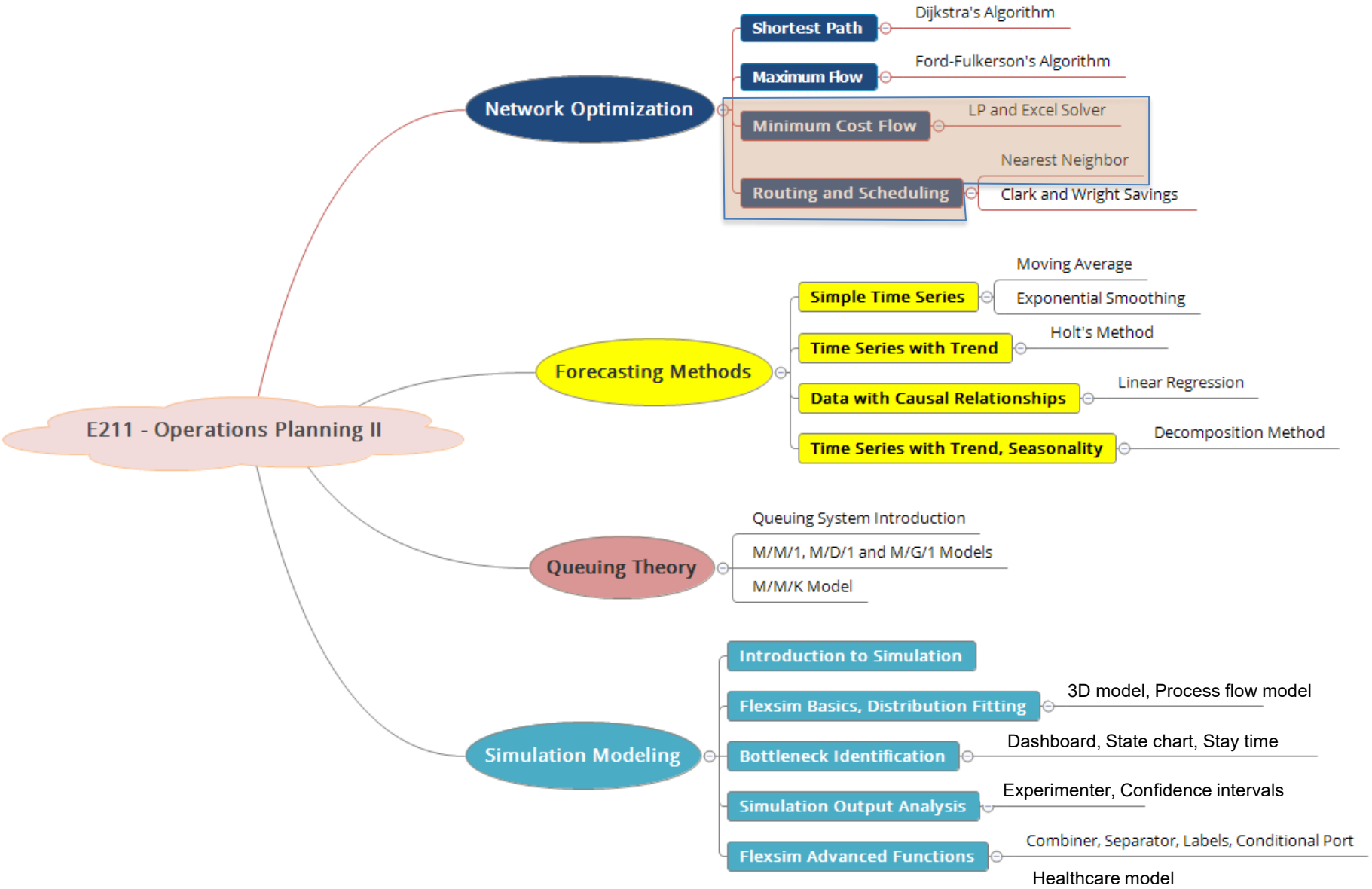
REPUBLIC POLYTECHNIC

Lesson 19

E211 – Operations Planning II

SCHOOL OF ENGINEERING

Module Coverage: E211 Topic Tree



Minimum Cost Flow Problem

Distribution Network Problem

- In a distribution network problem,
  - Goods are shipped from known supply points (factories, plants) to known demand points (customers, retail outlets), possibly through intermediate points (regional and or/field warehouses) known as the transhipment points.

  - The overall objective is to find the optimal distribution plan:

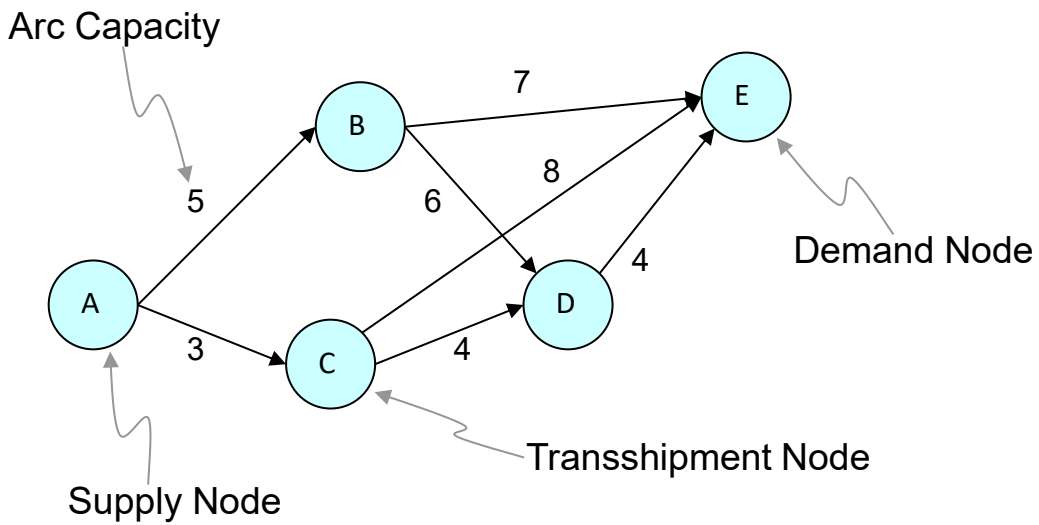
What is the **amount to ship along each route** from the supply points, through the intermediate points, to the demand points?

→ In order to **minimize the total shipping cost**.

Distribution Network Problem

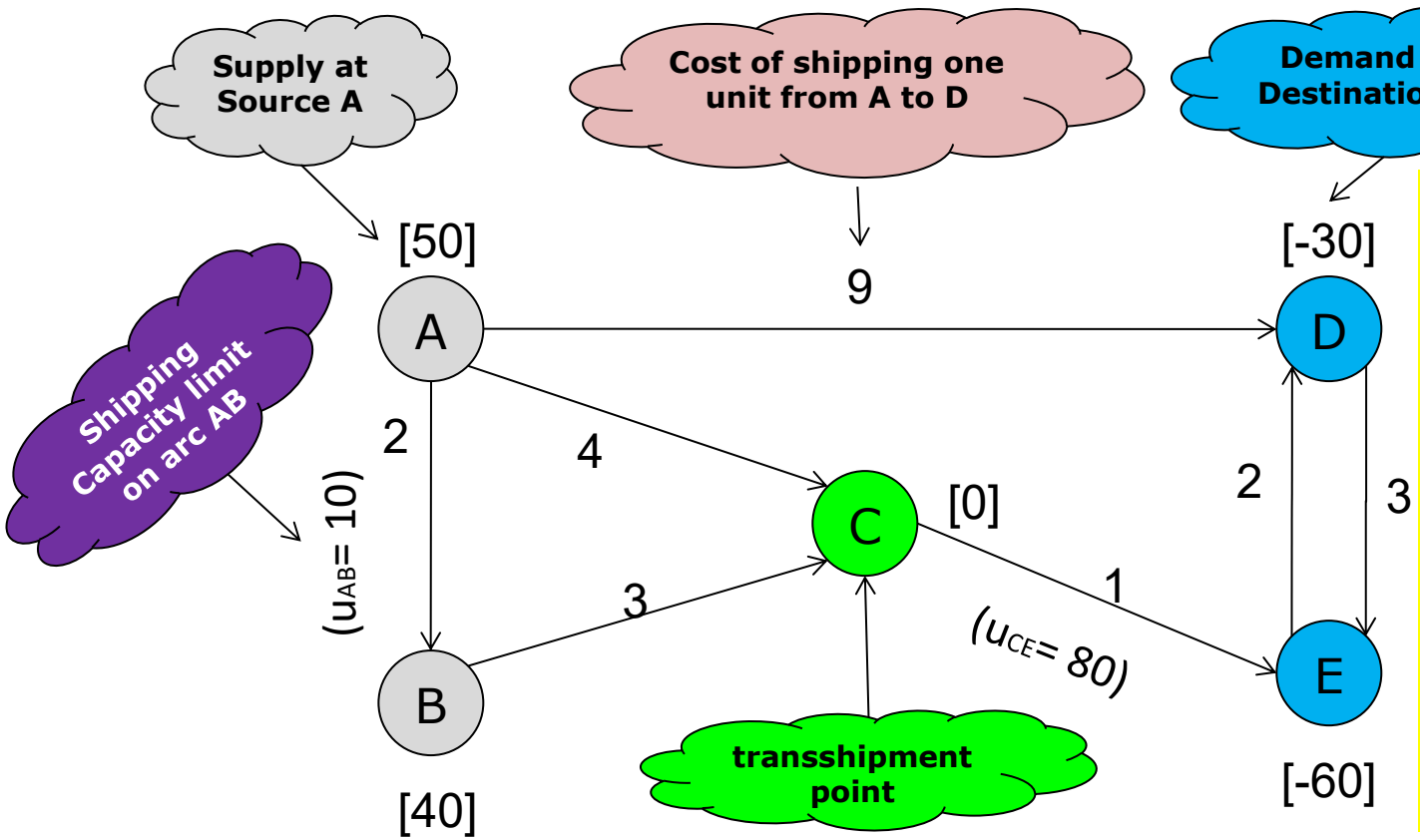
- In a distribution network problem,
  - Some or all of these constraints are present:
    - ✓ Not shipping more than the capacity from each supply point;
    - ✓ Meeting specified demand at the demand points;
    - ✓ Shipping goods only along valid routes;
    - ✓ Shipment not exceeding capacity limit imposed on certain routes.

Network Terminologies



- The maximum amount of flow that can be carried on a directed arc is referred to as the arc capacity.
- A supply node (or source node) has the property that the flow out of the node exceeds the flow into the node.
- The reverse case is a demand node (or sink node).
- A transshipment node (or intermediate node) satisfies the conservation of flow, so that flow in equals flow out.

Network Representation of Distribution Problem



Assumptions to simplify the problem:

- Homogenous goods
- Transportation costs are linear
- Shipping capacity is unlimited unless otherwise stated
- Supply and demand values are known and fixed

A and B are supply nodes: the flow out of the node exceeds the flow into the node. **Net flow generated > 0**

D and E are demand nodes: the flow into the node exceeds the flow out of the node. **Net flow generated < 0**

C is a transshipment node: flow in equals flow out. **Net flow generated = 0**

U_{AB} =10 represents upper bound (max) of flow capacity from A to B.

The Minimum Cost Flow Model

- The distribution network problem can be precisely represented and solved by the Minimum Cost Flow Model.

- The Minimum Cost Flow Model is an optimization model with the following features:
  - Its objective is to find the **cheapest** way of sending a certain amount of flow **from sources to destinations** through a flow network;
  - Like the <u>transportation problem</u>, it can consider multiple sources and multiple destinations for the flow, again with associated costs.

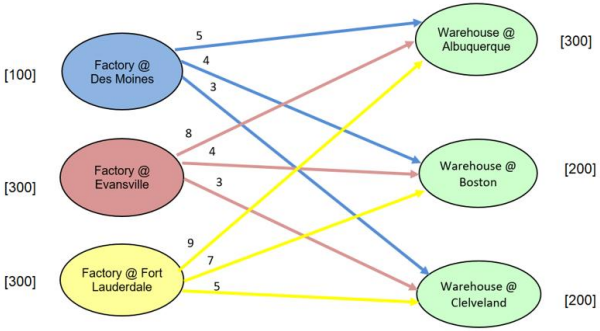
Special Case of Minimum Cost Flow Model

Two other types of problems are special cases of the Minimum Cost Flow Problem:

- **Maximum Flow Problem**: Aim is to maximize the amount of flow from an origin to a destination when the arcs of the network have limited flow capacities.
- Examples include flow of water/ gas/ oil through a network of pipelines or flow of traffic through a network of roads

- **Shortest Path Problem**: Aim is to find the smallest distance/ cost/ time between an origin and various destination points.
- For example, finding the best route to go from one town to another on a road map through a number of intermediate towns.

The Minimum Cost Flow Model



Transportation/ Assignment Problem

- No transhipment nodes
- No limitation on flow capacity
- Each arc is directed from a source node to a destination

Shortest Path Problem

-Single source (O) having supply of '1'

-Single sink (D) having demand of '1'

-No limitation on flow capacity

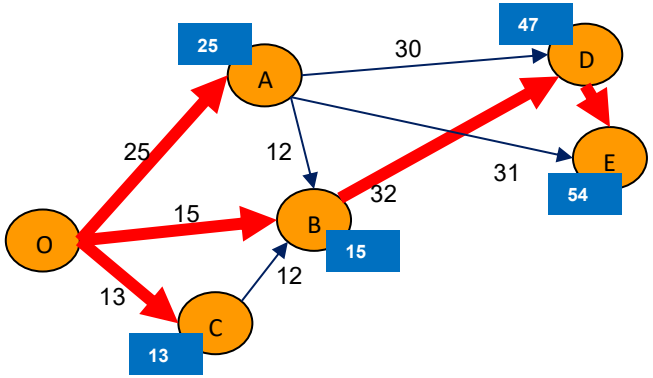
Minimum Cost Flow

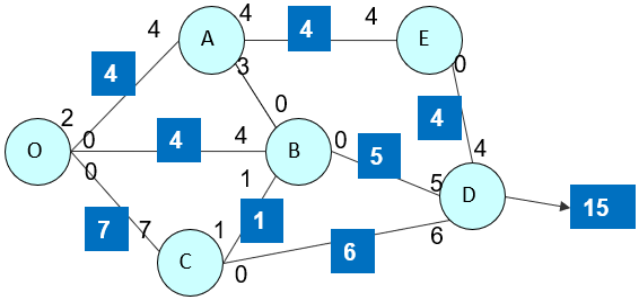
Maximum Flow Problem

-Single source (O); single sink (D)

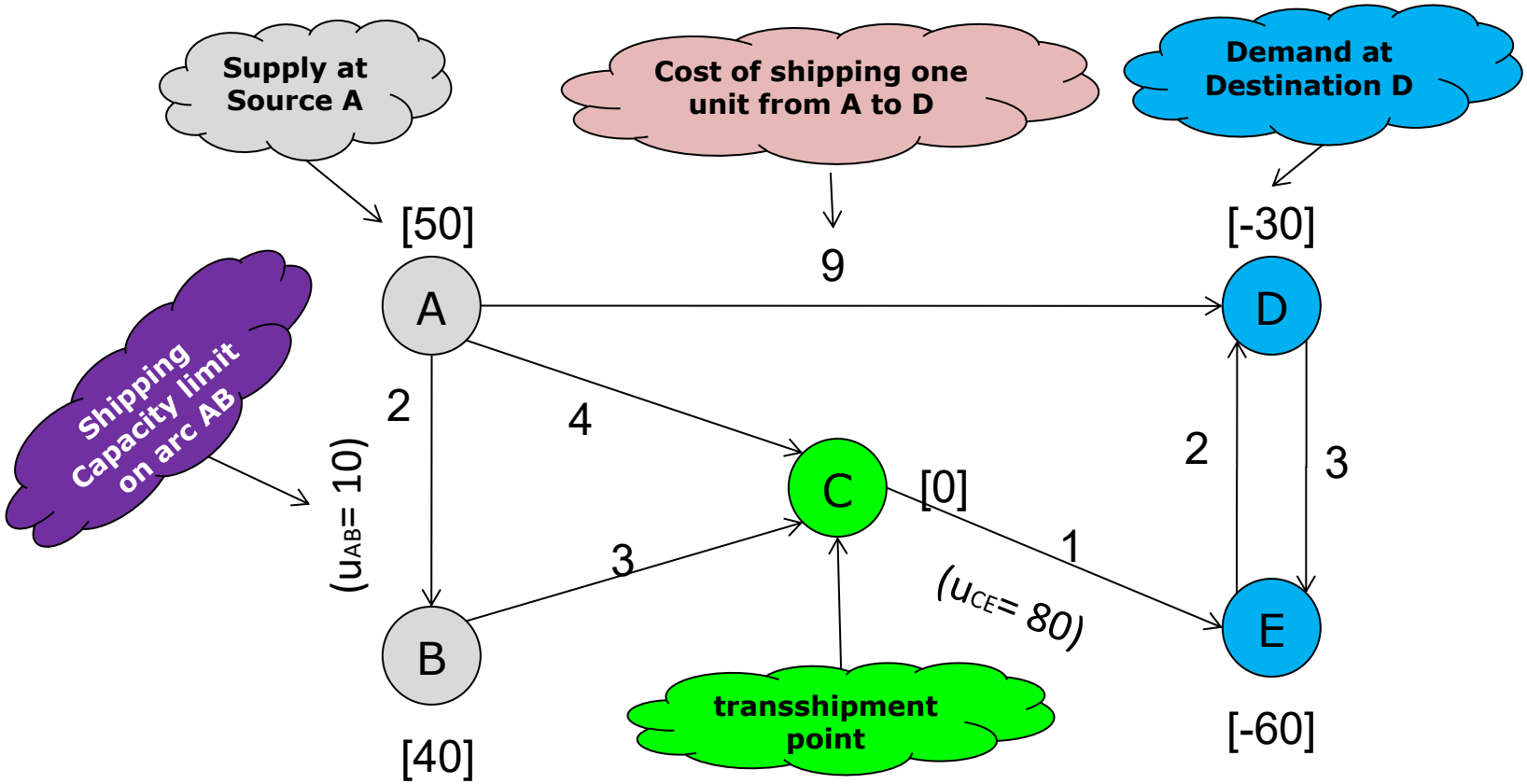
-With limitation on flow capacity

-No consideration on cost of flow





Network Representation of Distribution Problem



To minimize the total cost of shipment while satisfying the demands, how should you plan the distribution?

LP Formulation of the Minimum Cost Flow Problem

- **Decision Variables**: x_{ij} represents the amount of flow through arc $i \rightarrow j$ where i, j represents nodes in the network: $x_{AB}, x_{AC}, x_{AD}, x_{BC}, x_{CE}, x_{DE}, x_{ED}$
- **Objective Function:**

  Minimize total shipping cost, $Z = 2x_{AB} + 4x_{AC} + 9x_{AD} + 3x_{BC} + x_{CE} + 3x_{DE} + 2x_{ED}$
- **Constraints:**
  - **Flow constraints** (one for each node)

    Amount shipped out – Amount shipped in = Net flow generated at the node

    Node A: $x_{AB} + x_{AC} + x_{AD} = 50$

    Node B: $-x_{AB} + x_{BC} = 40$

    Node C: $-x_{AC} - x_{BC} + x_{CE} = 0$

    Node D: $x_{DE} - x_{AD} - x_{ED} = -30$

    Node E: $-x_{CE} - x_{DE} + x_{ED} = -60$
  - **Upper-bound constraints**

    Flow through an arc cannot exceed the limited arc capacity

    $x_{AB} \leq 10; \quad x_{CE} \leq 80.$
  - **Non-negativity constraints**: All $x_{ij} \geq 0$

Integer Solution Property

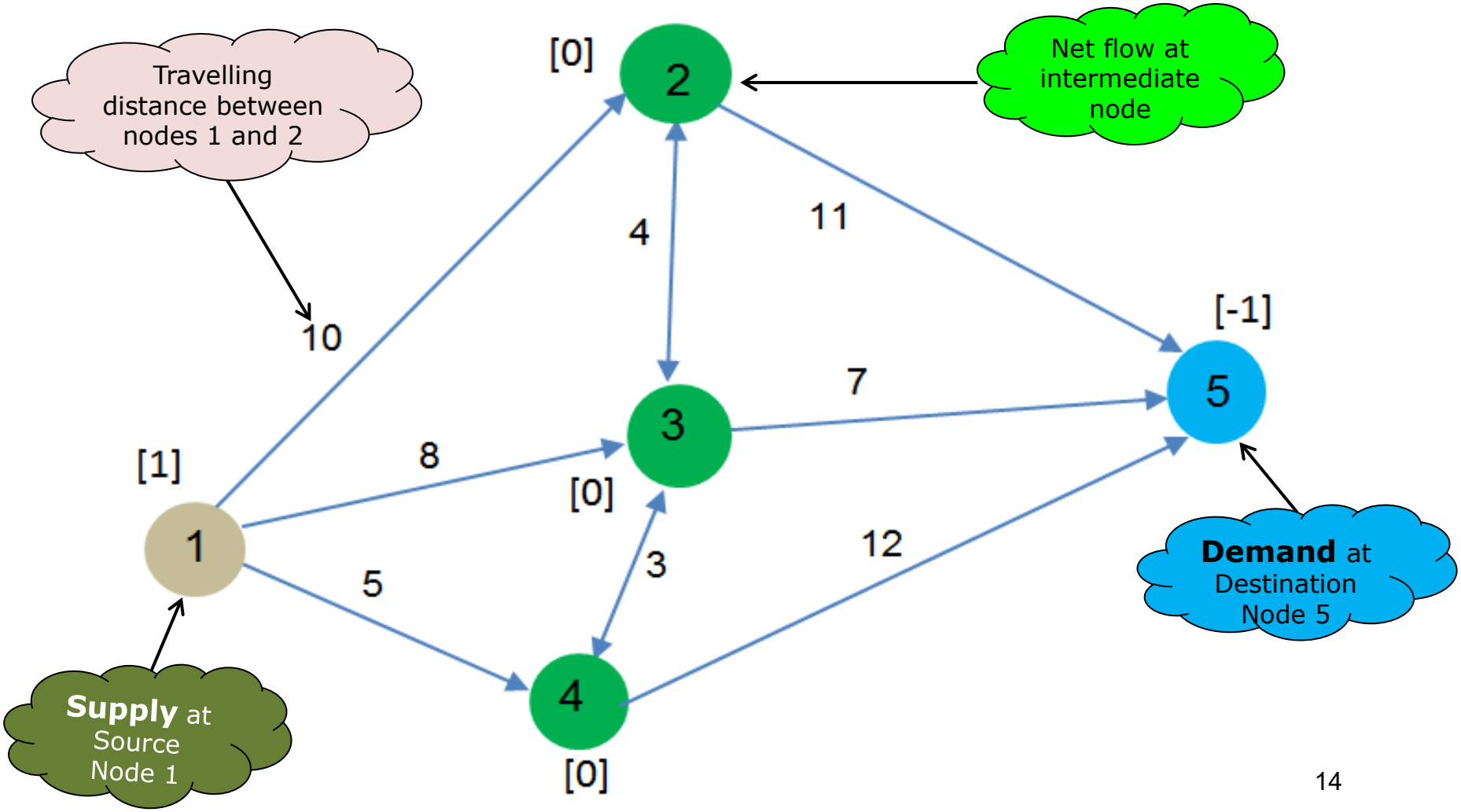
- Like the transportation problem, the minimum cost flow problem also has the integer solution property.

- <u>As long as every demand, supply and arc capacity have integer values</u>, a minimum cost flow problem with feasible solutions is guaranteed to have an optimal solution with integer values for all the decision variables.

- Therefore, it is not necessary to add constraints to the model to restrict the decision variables to only have integer values.

Applying Minimum Cost Flow Problem to the Shortest Path Problem

Finding the shortest path from Supply Node 1 to Demand Node 5.



LP Formulation for the Shortest Path Problem

Decision variables

Let X_{ij} be the route from node *i* to node *j*. X_{ij} is a binary variable:

$$X_{ij} = \begin{cases} 1 & \text{arc } i \rightarrow j \text{ is on the shortest path} \\ 0 & \text{otherwise} \end{cases}$$

In total, there are 10 decision variables: X_{12}, X_{13}, X_{14}, X_{23}, X_{25}, X_{32}, X_{34}, X_{35}, X_{43}, X_{45}

Objective function

Minimize distance traveled = $10X_{12} + 8X_{13} + 5X_{14} + 4X_{23} + 11X_{25} + 4X_{32}$
$+ 3X_{34} + 7X_{35} + 3X_{43} + 12X_{45}$

Ignore the decision variables going into source node and coming out of destination node.

LP Formulation for the Shortest Path Problem (with mandatory pass of each node)

Constraints

Subject to: -

Net flow contraints

Node 1: $x_{12} + x_{13} + x_{14} = 1$

Node 2: $-x_{12} - x_{32} + x_{23} + x_{25} = 0$

Node 3: $x_{32} + x_{34} + x_{35} - x_{13} - x_{23} - x_{43} = 0$

Node 4: $x_{43} + x_{45} - x_{14} - x_{34} = 0$

Node 5: $-x_{25} - x_{35} - x_{45} = -1$

Mandatory pass constraints

Node 2: $x_{12} + x_{32} >= 1$

Node 3: $x_{13} + x_{23} + x_{43} >= 1$

Node 4: $x_{14} + x_{34} >= 1$

Non-negativity constraints

All $X_{ij} >= 0$

- Net flow constraints already imposed mandatory pass on source node 1 and destination node 5.
- Mandatory pass constraints are required only for intermediate (transhipment) nodes 2, 3 and 4.
- Here we use incoming arcs to formulate the mandatory pass constraints. We can also use outgoing arcs to formulate them.
- Due to the 'integer solution' property of the minimum cost flow problem, we do not need to specify the decision variables to be binary.

For the shortest path problem with no mandatory pass requirement, we can ignore this set of constraints.

Learning Outcomes

For the Minimum Cost Flow Problem, students should be able to:

- Explain that a distribution network problem deals with the distribution of goods from ‘sources’ to ‘destinations’ in a network with transshipment points and may consider cost of shipment and flow capacity.
- Formulate LP models for the minimum cost flow problem and shortest path problem.
- Solve LP models using Excel Solver for the minimum cost flow problem and shortest path problem.

Delivery Routing Problem

Part 1

Routing and Scheduling Problems

- The scheduling of customer service and the routing of service vehicles are at the heart of many service operations.
  - Service routing: dispatching of installation or service technicians
  - Passenger routing: routing and scheduling of school buses, public buses, ride-sharing
  - Freight routing: pickup and delivery service of logistics firms; postal and parcel delivery
- **Scheduling** involves planning **the timing** for each location to be visited.
- **Routing** involves forming **the sequence** in which locations are to be visited.

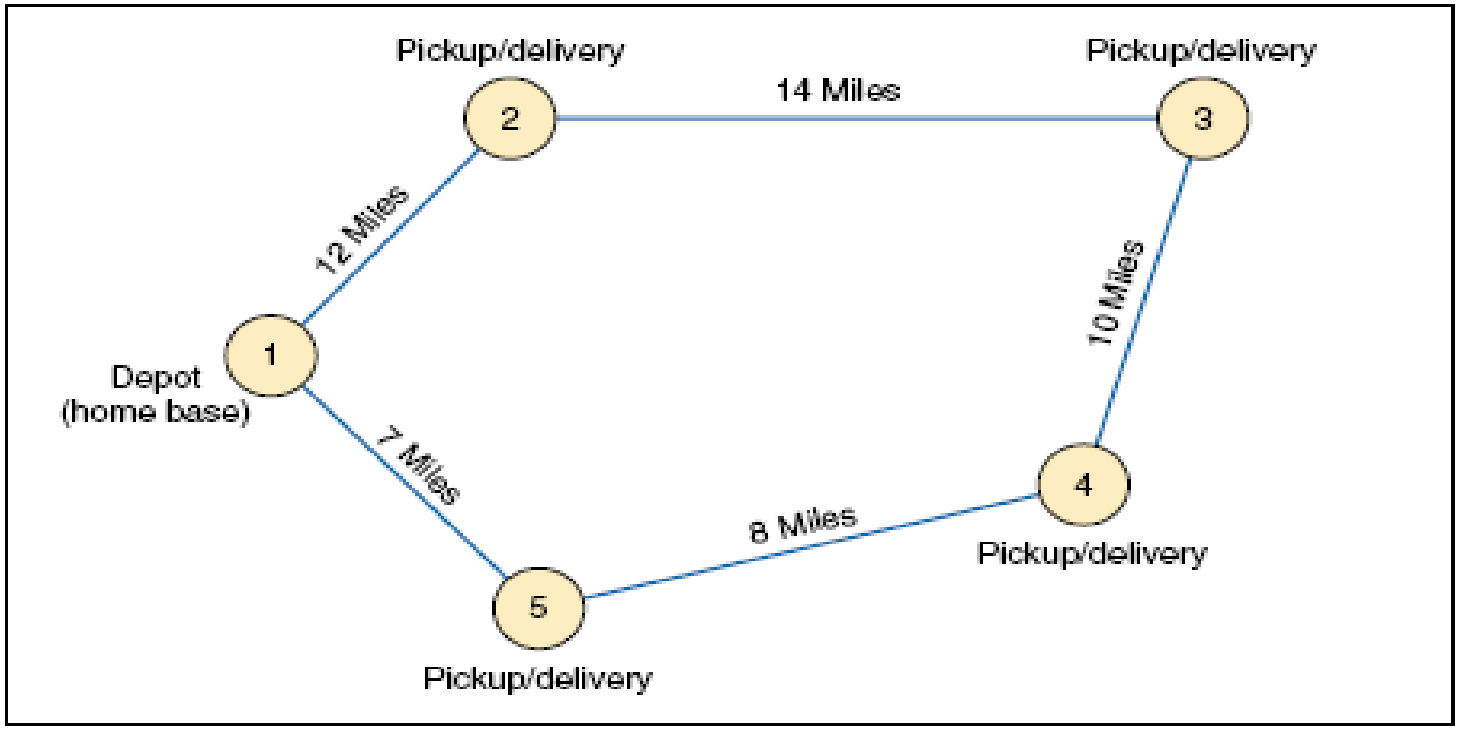
Objectives and Constraints of Routing and Scheduling Problems

- Minimize total cost of service delivery or route by one or more of the following:
  - Minimize total distance traveled
  - Minimize total time traveled
  - Minimize number of vehicles utilized
  - Minimize number of personnel on duty
- Subject to one of more constraints (not exhaustive):
  - Vehicles
    - ✓ Capacities
    - ✓ Release Time, Maximum Time and Down Time
  - Customers
    - ✓ Time windows (Timing open for receiving of goods)
    - ✓ Priority
    - ✓ Pickup and delivery

Characteristics of Routing and Scheduling Problems

- Can be represented using network diagrams, using nodes and arcs
  - Nodes represent locations to be visited. Depot node is the begin/end location
  - Arcs represent the possible paths among the locations. Arcs can be directed (one-way, represented by arrow) or undirected (two ways, represented by line)
- Schedule – time for each node to be visited
- Route – sequence in which nodes are to be visited

Feasibility of a Tour



Example of a tour for the above network: 1 → 2 → 3 → 4 → 5 → 1

For a tour to be feasible,

1. It must start and end with the depot node
2. All nodes must be included
3. A node must be visited only once

Classification of Routing and Scheduling Problems

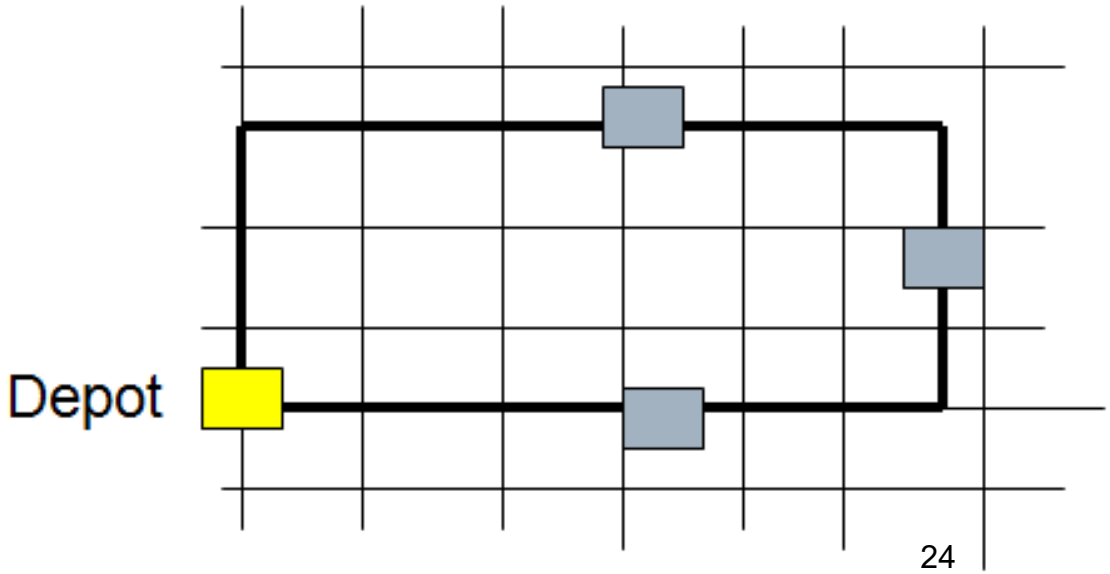
- Routing and scheduling problems can be classified into the following categories:
  - **Traveling salesman problem**
  - Multiple traveling salesman problem
  - **Vehicle routing problem**
  - Vehicle routing problem with time windows
  - Pickup and delivery problem with time windows

Travelling Salesman Problem

- Given a set of towns and the distances between them, the TSP determines the shortest tour starting from a given town, passing through all the other towns exactly once and returning to the first town.

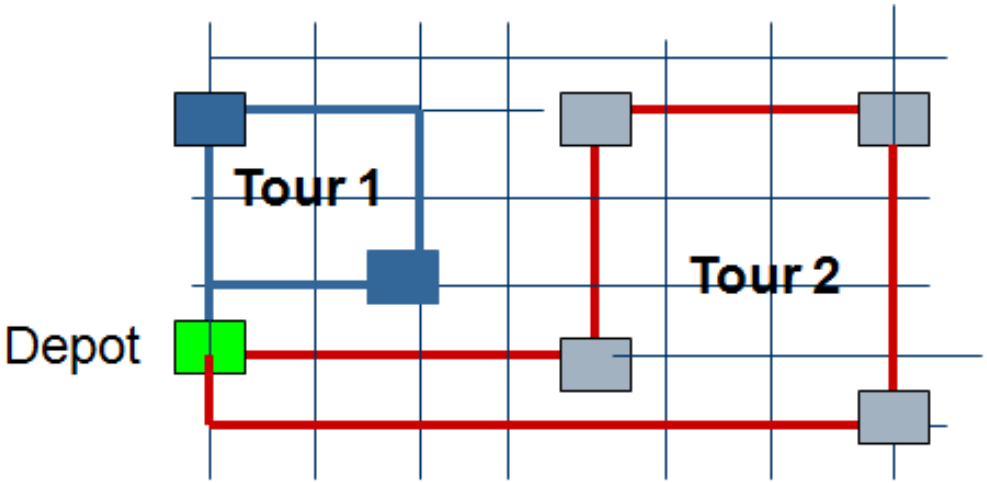
- Example - sales representative attending appointments at a number of different locations before returning to base.

- Type of decisions
  - Routing



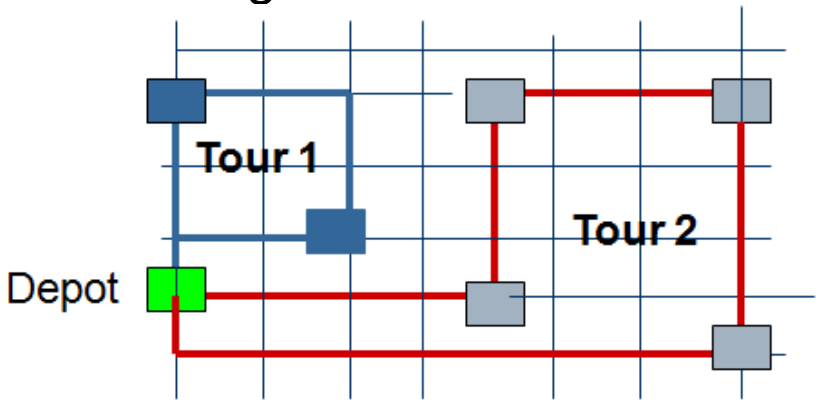
Multiple Travelling Salesman Problem

- The MTSP is a generalization of the TSP, where more than one salesman (vehicle) is needed to form the tours due to the limit on time (distance) travelled by each salesman.
- In MTSP, all the salesman (vehicle) are to leave from and return to a common depot.
- Typical applications of MTSP are in service routing
  - Example - dispatching of more than one installation/repair technician
- Type of decisions
  - ➢ Assigning
  - ➢ Routing



Vehicle Routing Problem

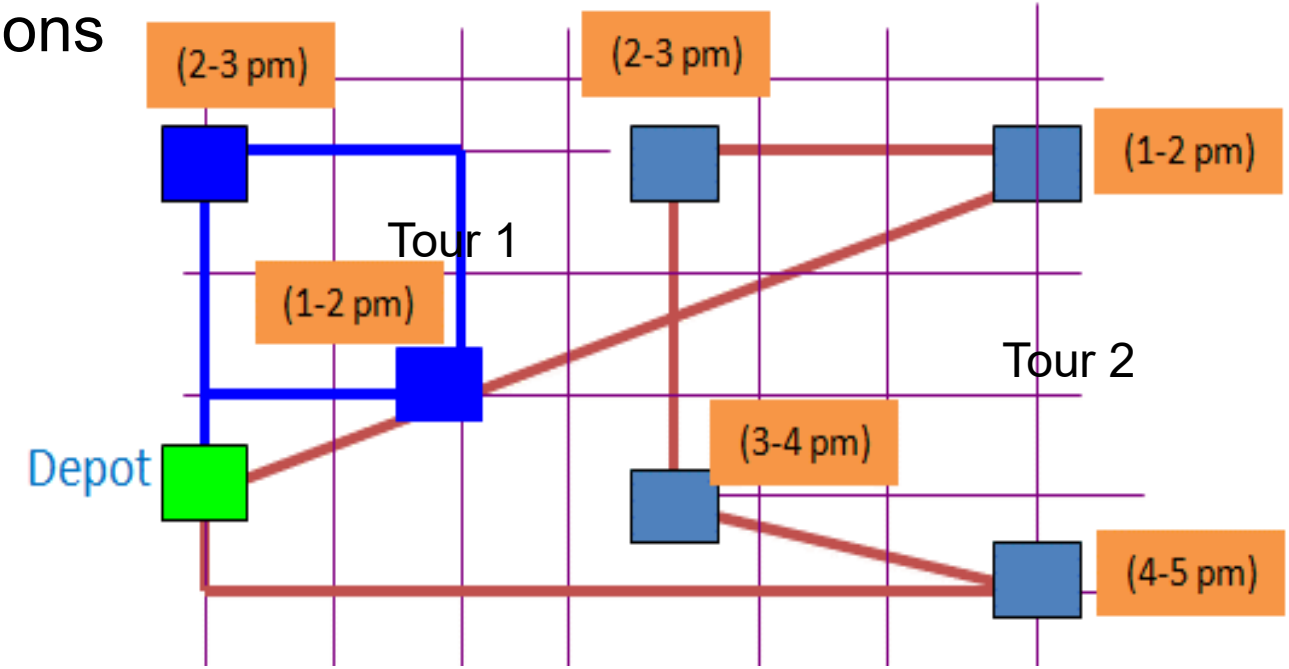
- In VRP, a fleet of vehicles located at a central depot has to serve a set of geographically dispersed customers and return to depot.
- Each vehicle has a given capacity and each customer has a given demand.
- The objective is to minimize the total cost (travelling distance) of serving the customers.
- Example - delivery of goods located at a central depot to customers who have placed orders for such goods.
- Type of decisions
  - Assigning
  - Routing



Vehicle Routing Problem with Time Windows

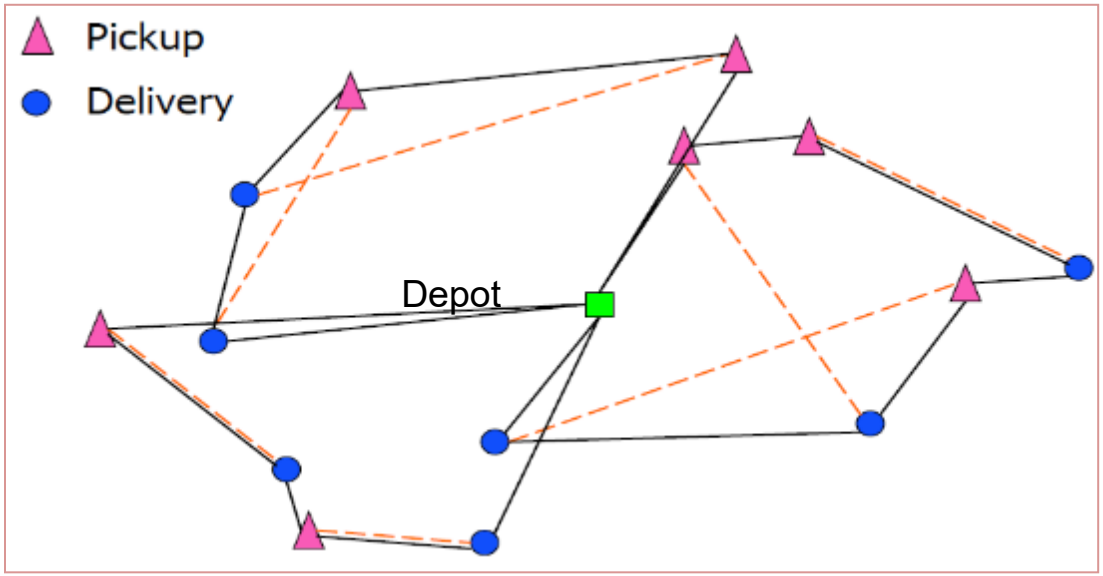
- The VRPTW is similar to the VRP except that, each customer may require to be served within a given time window.

- Example - Delivery of perishable food supplies
- Type of decisions
  - Assigning
  - Routing
  - Scheduling



Pickup and Delivery Problem with Time Windows

- The PDPTW is similar to the VRP except that, in addition to the size of the load to be transported, each customer service also specifies
  - The location where the load is to be picked up and the pickup time window;
  - The location where the load is to be delivered and the delivery time window;
- Example –
  - Transfer of stocks among retail stores
- Type of decisions
  - Assigning
  - Routing
  - Scheduling



The PDPTW determines how to form the routes and schedule the pickup and delivery.

Solution approaches to the Routing and Scheduling Problems

- Exact algorithms such as integer programming algorithms
  - For a problem with *n* drop-off points, there are *n*! possible routings. E.g. 8 drop-off points → 40320 possible routings!
  - The solution space increases extremely fast as the number of drop-off points increases, and therefore exact algorithms are often not suitable for large size problems.

- **Heuristic algorithms**
  - Rule-of-thumb methods
  - May not achieve optimality but solutions are often reasonably good.

- Meta-heuristics
  - Top-level heuristics guiding other heuristics to search for better feasible solutions in the solution space, such as Genetic Algorithms, Ant Colony Optimization, Simulated Annealing, Tabu search.
  - They are very commonly used in solving complex vehicle routing and scheduling problems.

Solution approaches to the Routing and Scheduling Problems

- Commonly used vehicle routing heuristics
  - **Construction algorithm:** an algorithm that determines a tour according to some construction rules. Examples are **Nearest Neighbor method and Clarke and Wright Savings method.**
  - **Improvement algorithm:** an algorithm that performs a sequence of arc or node exchanges within or between vehicle routes to improve the current tour.
  - **Two-phase algorithm:** an algorithm that constructs vehicle routes in two phases:
    - ✓ **Cluster-first-route-second method:** customers are first organized into feasible clusters, and a vehicle route is constructed for each of the clusters.
    - ✓ **Route-first-cluster-second method:** a tour is first built on all customers and the tour is then segmented into feasible vehicle routes.

Tour construction—Nearest Neighbor Method

- Select a node 1 (depot) to start with
- Add the closest node to node 1
- Repeat by adding to the last node the closest unvisited node until no more nodes are available
- Connect the last node with the first node

Nearest Neighbour Method – An Example

Distance Table

From	To (Distance in km)						
	A (Depot)	B	C	D	E	F	G
A (Depot)	0	22	19	15	7	12	26
B	22	0	5	16	18	12	30
C	19	5	0	6	14	8	16
D	15	16	6	0	8	3	21
E	7	18	14	8	0	6	25
F	12	12	8	3	6	0	28
G	26	30	16	21	25	28	0

Starting from A (Depot), nearest node is E.
From E, the nearest node is F.
From F, the nearest node is D.
And so on…

Tour determined using Nearest Neighbour method:
A -> E -> F -> D -> C -> B -> G -> A
Total distance = 7+6+3+6+5+30+26 = 83 km

Learning Outcomes

For the Delivery Routing Problem, students should be able to:

- Identify what is a Vehicle Routing and Scheduling problem: Objectives, constraints, characteristics and classification of problems.
- Apply heuristic algorithm: Nearest Neighbour Method to obtain a feasible route.